

Cooper Braun

360-900-3092 | cooperjbraun13@gmail.com | [linkedin.com/in/cooper-braun-gu](https://www.linkedin.com/in/cooper-braun-gu) | github.com/cooperbraun13

EDUCATION

Gonzaga University

Bachelor of Science in Computer Science

Aug 2023 – May 2027

GPA: 3.53 / 4.0

Relevant Coursework

Data Structures and Algorithms, Computer Security, Database Management Systems, Linux & DevOps

EXPERIENCE

Undergraduate Research Assistant

Gonzaga University Institute for Informatics & Applied Technology

Sep 2025 – Present

Spokane, WA

- Built a bracket analysis tool in Python for Hoopfest (the world's largest 3-on-3 basketball tournament), ingesting and joining multi-table CSV data across 20,000+ players and 500+ brackets to evaluate team cohesion across age, height, and experience dimensions
- Designed a composite fit-scoring system that quantifies how well each team belongs in its bracket relative to the broader field, surfacing 750+ ranked swap suggestions and flagging outliers at both the player and team level
- Generated self-contained interactive HTML dashboards from script output, enabling non-technical tournament staff to explore flagged teams and compare bracket placements without additional tooling

Software Engineering Intern

InterRoom

Jun 2025 – Aug 2025

New York, NY

- Built a Flask backend wrapping a multi-channel outreach API, enabling automated delivery of emails, LinkedIn webhooks, and Reddit posts triggered from a single internal interface
- Designed and deployed a dashboard over the outreach API using Flask and PostgreSQL, allowing non-technical users to trigger and monitor outreach campaigns without direct script access
- Containerized services with Docker and implemented GitHub Actions CI/CD pipelines, improving system uptime from 85% to 99% and eliminating manual deployment steps

PROJECTS

BluPrnt | *GitHub Repo*

TypeScript | Express | PostgreSQL | Docker

- Led a team of 5 to build a contractor-vendor comparison platform from concept to MVP, owning architecture decisions and the full implementation lifecycle from schema design through deployment
- Designed a normalized 6-table PostgreSQL schema and wrote efficient SQL queries to enforce relational data integrity and support high-performance lookups
- Built 20+ RESTful backend routes in Express and TypeScript supporting dynamic filtering, ranking, and search across all core resources, with strict input validation via Zod

Tapo Tracker | *GitHub Repo*

Next.js | TypeScript | Supabase | Vercel

- Designed and implemented the full data layer in Supabase - relational schema, cascading foreign keys, and a centralized query module as the sole point of database access across the application
- Built a deterministic scoring engine in TypeScript implementing a multi-criteria tiebreaker chain (points, perfect picks, correct picks) with pot-splitting logic for unresolved ties
- Implemented invite-only authentication using Supabase Auth, including two invite flows (claim existing record vs. create new), PKCE and OTP callback handling, and role-based route protection via Next.js middleware
- Deployed to Vercel with GitHub Actions CI, keeping all derived stats computed at runtime from raw stored scores rather than persisted state

TECHNICAL SKILLS

Languages: Python, SQL, Go, Java, C/C++, JavaScript, TypeScript

Frameworks & Libraries: Flask, Express, Node.js, Next.js, Zod, Pandas, NumPy, Scikit-learn, Matplotlib

Tools & Platforms: Docker, Linux, GitHub Actions, GCP, AWS, Supabase, Prisma, JWT, OAuth, Postman